LAB WORK.

Lab week - 8

Huffman coding

" Data Analytics and Intelligence lab coding"

" DATA ANALYTICS INTELLIGENCE LABORATORY"

Frequency :	Letter	Count
	A	7
	L	4
	N	4
	E	3
	I	3
	C	2
	D	2
	O	2
	R	2
	Y	2
	B	1
	G	1
	S	1

B	G	S	C	D	O	R	Y	E	I
1	1	1	2	2	2	2	2	3	3

L	N	T	A
4	4	4	7

~~Applying huffman algorithm.~~

~~B [1] - G [1] - S [1] - [2] - D [2] - O [2] [2] - Y [2] - C [3] - I [3] - E [4] - N [4] - T [4] - A [7]~~

Applying huffman algorithm

B[1] - G[1] - S[1] - C[2] - D[2] - R[2] - Y[2]
- E[3] - I[3] - L[4] - N[4] - T[4] - A[7]

I - 1 Combine B[1] and G[1] to get BG[2]
S[1] - C[2] - D[2] - R[2] - Y[2] - BG[2]
E[3] - I[3] - L[4] - N[4] - T[4] - A[7]

I - 2 Combine S[1] and C[2] to get SC[3]
D[2] - D[2] - R[2] - Y[2] - BG[2] -
E[3] - I[3] SC[3] - L[4] - N[4] - T[4] - A[7]

I - 3 Combine D[2] and D[2] to get DD[4]
R[2] - Y[2] - BG[2] - E[3] - I[3] - SC[3]
- L[4] - N[4] - T[4] - DD[4] - A[7]

I - 4 : Combine R[2] and Y[2] to get RY[4]
BG[2] - E[3] - I[3] - SC[3] - L[4]
N[4] - T[4] - DD[4] - RY[4] - A[7]

I - 5 Combine BG[2] and E[3] to get BGE[5]
I[3] - SC[3] - L[4] - N[4] - T[4] - DD
[4] - RY[4] - BGE[5] - A[7]

I - 6 Combine I[3] SC[3] to get ISC[6]
L[4] - N[4] - T[4] - DD[4] RY[4] - BGE[5]
ISC[6] - A[7]

I - 7 Combine L[4] and N[4] to get LN[8]
T[4] - DD[4] - RY[4] - BGE[5] - ISC[6]
- A[7] - LN[8]

I 8 : Combine T [4] DO [4] to get TD [8]
Ry [4] - DGE [5] - IS [6] - A [7] - LN [8] - TDO [8]

I 9 : Combine : Ry [4] and DGE [5] to get Ry DGE [9]

ISC [6] - A [7] - LN [8] - TDO [8] - Ry DGE [9]

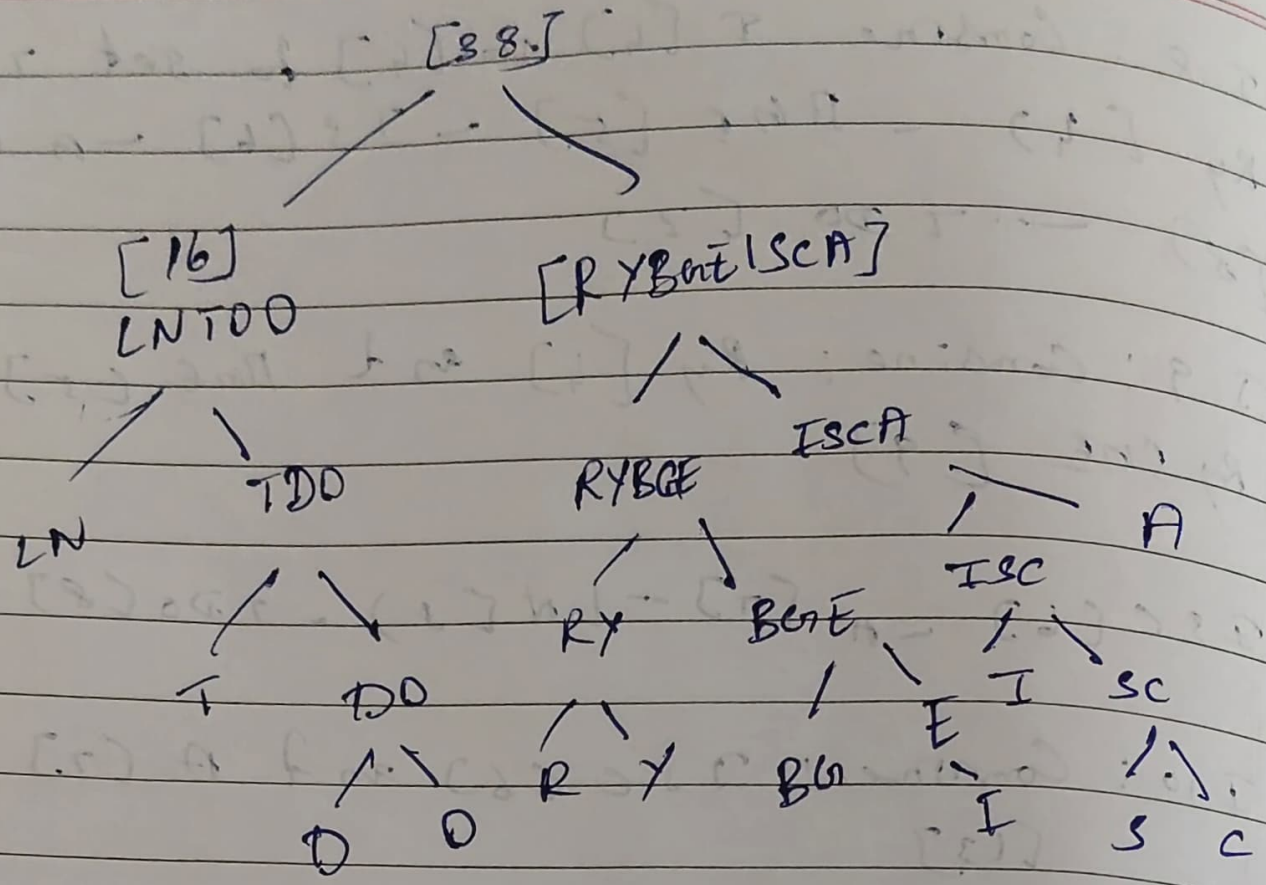
I 10 : Combine : ISC [6] and A [7] to get ISCA [13]

LN [8] TDO [8] - Ry DGE [9] - ISAC [13]

I 11 : Combine : LN [8] 2 TDO [8] to get
LNTDO [16] Ry DGE [9] - ISCA [13] - LN TDO [16].

I 12 : Combine Ry DGE [9] 2 ISCA [13] to get
Ry DGE ISCA [22]
LNTDO [16] Ry DGE ISCA [13] to get
~~Ry DGE ISCA~~ [22]
~~LNTDO~~

I 13 : LN TDO Ry DGE ISCA [38]



A = 11 ; L = 000 ; N = 001 ; T = 010 ; E = 1011 ;

I = 1100 ; C = 11011 ; D = 0110 ; O = 0111

R = 1000 ; Y = 1001 ; B = 10100 ; G = 10101 ; S = 11010